Emulating Cuttlefish Behavior and Intelligence in a Novel Robot

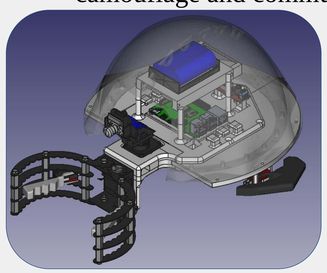
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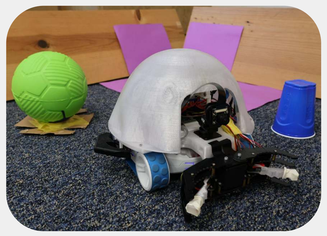
Abstract/Objectives

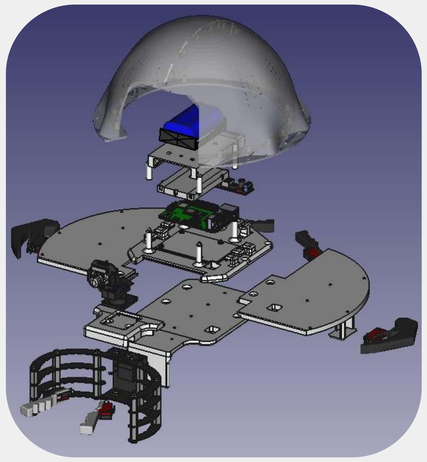
- **Project Goal:** Create a biomimetic robot, the CuttleBot, that embodies the cognitive abilities and behavioral intricacies of cuttlefish, leveraging the expertise of a diverse team from multiple disciplines
- **Background:** Recent research highlights the cuttlefish's advanced cognitive skills, such as rapid learning, memory retention, and problem-solving abilities in complex tasks previously only tested on vertebrates
- **Contributions:**
  - **Adaptive Camouflage:** Implements camouflaging techniques in response to distant predators, enhancing the robot's ability to blend with its environment
  - **Immediate Defense Mechanisms:** Features blanching and hiding behaviors when predators are too close, mimicking cuttlefish's rapid response to threats
  - **Confusion Tactics:** Employs a confusion signal using its shell just before launching attacks, which aids in distracting and capturing prey effectively
  - **Dynamic Learning:** Utilizes Q-learning to adaptively respond to varying populations of predators and prey, improving its survival and hunting strategies over time

Materials

- **Hardware Configuration and Control:**
  - **Robotic Base and Control:** The CuttleBot is built on a Sphero RVR+ and paired with a Raspberry Pi 4B for central processing
  - **Visual Processing:** A high-resolution, wide-angle camera mounted on a dual-axis pan-tilt unit, providing a comprehensive 160-degree field of view
- **Sensory Integration and Feedback Mechanisms:**
  - **Tactile Feedback:** Installation of limit switches on the claw and along the robot's body to detect physical contact and barrier interactions.
  - **Adaptive Display Shell:** Custom-designed, 3D-printed semi-transparent shell integrated with programmable LED arrays for rapid color and pattern changes, simulating cuttlefish skin camouflage and communication signals





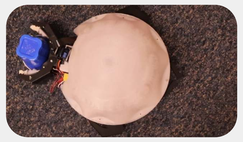


Methods

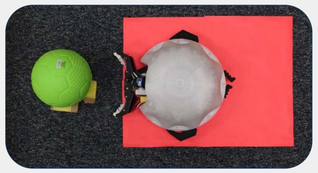
- **Behavioral and Learning Algorithms:**
  - **Q-learning Implementation:** Employs a model-free reinforcement learning algorithm to dynamically adapt behaviors based on environmental feedback, differentiating between predator and prey encounters
  - **Behavioral Phases:**
    - **Attack Phase**: Involves object recognition, alignment, and striking. Enhanced by employing a hypnosis pattern to distract and effectively capture prey

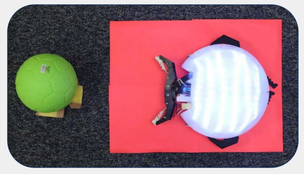


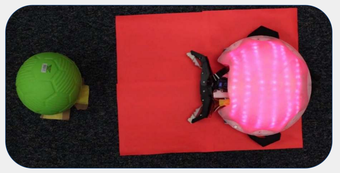




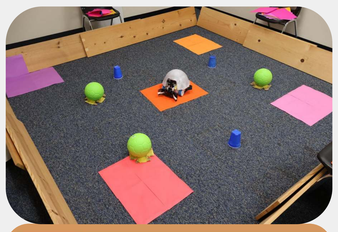
    - **Defense Phase:** Begins with the robot employing its defense mechanisms of blanching and retreating. As it learns, it activates its camouflage abilities from a distance when it recognizes a predator as a threat
      - **Blanch:** The blanching action includes flashing a white color and jumping backwards
      - **Camouflage:** The Cuttlebot's shell matches the color of the environment using a color sensor and color conversion calculations

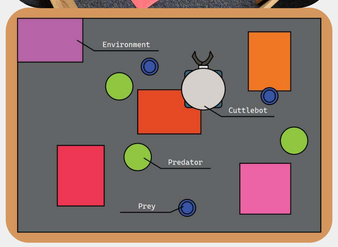






- **Experimental Environment and Testing:**
  - **Arena Configuration:** An enclosed arena equipped with sensors and variable obstacles, hosting different numbers of green ball "predators" and blue cup "prey"
  - **Testing Protocol:** Each scenario undergoes five trials to ensure data reliability, adjusting variables like predator/prey ratios, learning rates, and beta values



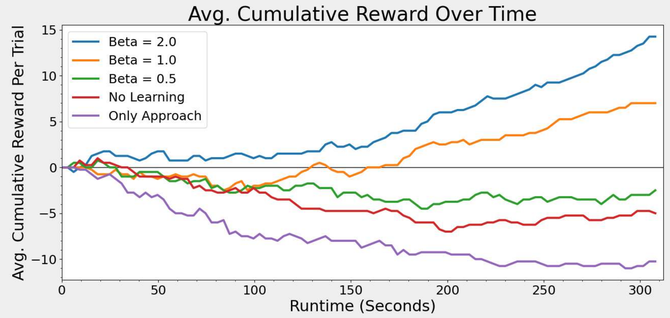


Testing Conditions

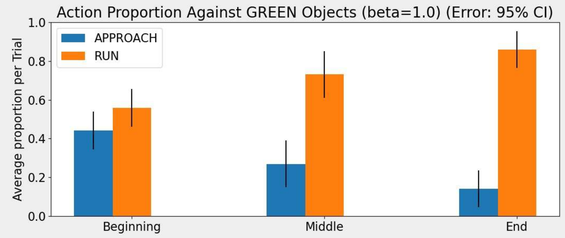
ROBOT CONDITIONS	LR=0.0	LR=0.1	LR=0.2	LR=0.4
β=0.0	ONLY APPROACH	N/A	N/A	N/A
β=0.5	N/A	N/A	EXPLORATIVE BEHAVIOR	N/A
β=1.0	NO LEARNING	LOW LEARNING RATE	BASELINE BEHAVIOR	HIGH LEARNING RATE
β=2.0	N/A	N/A	EXPLOITATIVE BEHAVIOR	N/A

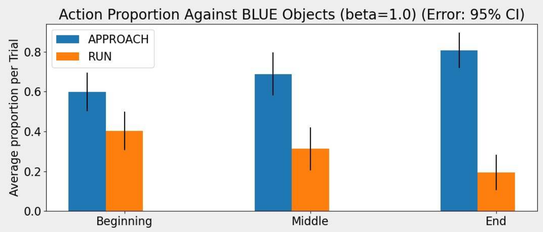
ENVIRONMENT CONDITIONS	PREY=1	PREY=3
PREDATORS=1	SPARSE ENVIRONMENT	OVERWHELMING PREY
PREDATORS=3	OVERWHELMING PREDATORS	ABUNDANT ENVIRONMENT

Results and Discussion



- **Adaptive Behavior to Maximize Rewards:** The CuttleBot achieved optimal learning with a beta value of 1.0, balancing rapid adaptation and effective behavior exploitation. In contrast, a beta of 0.5 delayed learning, while a beta of 2.0 led to excessive exploitation, compromising the robot's ability to adapt flexibly





- **Learned Behavior:** To maximize the reward received, the CuttleBot learned overtime to run from green objects (i.e. predators) and approach blue objects (i.e. prey)

References

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